Fiscal Year:	FY 2022	**Task Last Updated:**	FY 12/22/2021
PI Name:	Bell, Suzanne Ph.D.		
Project Title:	Human Factors and Behavioral Performance Exploration Measures in HERA		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	03/01/2021	**End Date:**	01/31/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	December 2021 report: Sheena I. Dev, PhD, was added as co-Investigator in May 2021.		
COI Name (Institution):	Whiting, Sara Ph.D. (BHP Laboratory, KBR/NASA Johnson Space Center) Landon, Lauren Ph.D. (BHP Laboratory, KBR/NASA Johnson Space Center) Dev, Sheena Ph.D. (BHP Laboratory, KBR/NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
	NOTE (9/30/2021): Continuation of "Human Factors and Behavioral Performance Exploration Measures in HERA (previous title: Identification and Validation of BHP Standard Measures in HERA for Transport)"--PI Dr. Peter Roma, who moved to Naval Research Laboratory (NRL) in early 2021. With Dr. Roma's move to the NRL in early 2021, Dr. Suzanne Bell is now Principal Investigator. NASA's Human Research Program (HRP) is developing a set of "Standard Measures" for use in spaceflight and analogs. These "standard measures" need to be validated in analog settings to ensure their validity, acceptability, and reliability in helping identify and measure indicators of risk associated with crew health and performance. Exploration missions are anticipated to create isolated and confined environments that will include stressors such as small teams living and working in extreme conditions for prolonged periods separated from family, friends; loss of the day/light		

Task Description:	cycle; loss or delay of communications with ground; partial gravity; and limited space, privacy, and food selection. HRP's Human Factors and Behavioral Performance (HFBP) Element seeks to maintain and enhance behavioral health and performance in such environments. The Behavioral Medicine risk (Risk of Adverse Behavioral or Behavioral Conditions and Psychiatric Disorders) is a high priority within the HRP because psychological and behavioral health factors reveal intra- and inter-individual variability. This variability poses significant challenge for measurement of risk and resilience in crews and individual astronauts during long-duration missions. The HFBP-EM (Exploration Measures) will build upon the initial use and identification of Behavioral Core Measures (BCM), whose purpose was to evaluate the feasibility and acceptability of these outcome measures (Dinges; NNJ13ZSA002N-BMED; project title: "Standardized Behavioral Measures for Detecting Behavioral Health Risks during Exploration Missions") in high fidelity analogs. The task proposed here provides an opportunity to continue using standardized measures in order to operationalize their efficiency and effectiveness, ensure consistent measurement of these key metrics, and facilitate the analysis and understanding of these measures in assessing risk. The development and use of these expanded standardized measures are needed now to develop astronaut norms in order to determine their efficacy and ability to objectively detect, assess, and manage off nominal events; and predict future off nominal events that may compromise a mission and increase the risk of behavioral health and performance decrements. This advances our need for individualized, real-time tools that provide rapid feedback and assessment that are needed for exploration class missions. The use of BHP (Behavioral Health and Performance) Standard Measures across analogs and analog missions, allows NASA Human Research Program (HRP) to determine a baseline for assessing and monitoring specific countermeasures; and increases the construct and predictive validity of countermeasures for assessing risk and for their monitoring of behavioral health and performance in spaceflight. Specific Aims include: Aim 1. Provide a set of BHP standard measurements for investigators to use in proposed projects. Significance: This allows NASA HRP to streamline and make more efficient the use of multiple measurements collected on research participants during analog research and helps to ensure a reduced burden on these research participants by using a "standard" set of measures for data sharing by multiple principal investigators (PIs). Aim 2. Enable comparison of multiple missions across spaceflight analog campaigns to quantify risk using reliable metric-based data. Significance: The "standardized measures" increase the generalization of findings across research analogs, increasing the validity and reliability of measures used to quantify, characterize, and assess the impact of spacelike analogs on prevalence of behavioral health issues, incidence rates, longer term health, and performance errors. This allows generation of reliable and sensitive metrics that can be used systematically to inform accurate risk assessment and mitigation status for future exploration spaceflight missions. Aim 3. Provide database for data-mining and integrative modeling and increase research data quality and transfer to the Life Sciences Data Archive (LSDA). Significance: Collection of "Standardized Measures" allows for greater consistency and fidelity of data collected, enhancing the data archiving capabilities of analog outcomes, increasing accessibility of data for NASA use, metrics assessment associated with red to yellow, and yellow to green risk status, and trending, and increasing the probability that the standardized set of measures are appropriately archived in a timely manner. Although data management agreements require PIs to submit their data after completion of their research, the exact format and timeliness of that data submission varies greatly among PIs. The collection of "standardized measures" via internal directed studies helps to ensure more timely, valid, and accessible data resources to help guide risk reduction. _____ NOTES 2018 August: Original Principal Investigator (PI) was Thomas J. Williams, PhD, who became HRP Human Factors & Behavioral Performance (HFBP) Element Scientist shortly after the award was made; Jason S. Schneiderman, Ph.D. served as interim PI until July 10, 2017, when Peter G. Roma, Ph.D. came to Johnson Space Center and became Principal Investigator. At that time, title was changed to "Identification and Validation of BHP Standard Measures in HERA for Transport" (original title was "Identification and Validation of BHP Standard Measures"). 2020 February: Project title was changed to "Human Factors and Behavioral Performance Exploration Measures in HERA" (Short Title: HFBP-EM in HERA).
Rationale for HRP Directed Research:	This research is directed because NASA must define complete scientific activities in a short time and there is insufficient time to issue a solicitation. This task is in direct response to both the August 2014 Human Subjects Review Board (HSRB) requirement and the baselined Human Research Program (HRP) Path to Risk Reduction milestone of providing standard measures to monitor crew health and performance. This will allow HRP to establish, evaluate, and manage a common set of measures for use in spaceflight and analog research to: develop baselines, systematically characterize risk likelihood and consequences, and assess effectiveness of countermeasures that work for human factors and behavioral performance risk factors. This proposal qualifies for a directed study due to the urgent, time-sensitive need to provide "standard measures" as the foundation to achieve consistent research measure for data-sharing in HERA (Human Exploration Research Analog) and to meet the highly constrained, operationally-focused data gathering and analysis that allows for greater consistency in the research methods that are specific to NASA HRP standard measures development. Additionally, the set of Behavioral Health and Performance (BHP) standardized measures in the HERA analogy reflects the more operational nature of the measures while allowing the multiple and frequent internal and external collaborations required to execute this study. The directed nature of this study also allows NASA to provide the unique research and support expertise that is needed to integrate and manage the data from all of the various participating studies to achieve HRP's intent and support to the Flight Analogs Project (FAP) within a highly constrained time schedule. Completion of the required research, support to both analog and operational requirements, and vetting of the evidence-based standards makes the solicitation process prohibitive. In addition, our group, the BHP Lab has already performed similar work across various analogs in support of previous NASA research and the use of the BHP Lab offers considerable efficiencies that are realized by building upon our existing work and that expertise. Finally, two panels of extramural subject matter experts with experience in critical task identification in academic, military, and spaceflight analog settings met at Johnson Space Center (JSC) and provided recommendations consistent with the enclosed project related to team processes (November 2015) and psychometric assessment of cognition (April 2016).
Research Impact/Earth Benefits:	

Task Progress:	Over the past year, the BHP Laboratory has made tremendous progress in three primary areas: (1) processing of previously collected data from HERA (Human Exploration Research Analog) Campaigns 4 and 5; (2) development of shorter forms of several of the HFBP-EM survey measures; and (3) protocol development and collection for HERA Campaign 6 (C6). First, we have made significant progress in the data cleaning, processing, and analysis of the HFBP-EM from HERA Campaigns 4 and 5. Careless responses on surveys were identified and flagged. Data from surveys which were originally developed and validated with Likert response formats but had been collected on visual analog scales were transformed and combined into validated subscales where applicable. Actigraphy and heart rate data were processed according to best practices, and sleep and heart rate metrics were summarized as daily aggregates. Objective performance tasks were subjected to data transformations and key performance metrics were extracted. A set of meta-data variables (e.g., mission, participant ID, mission day) were generated to define key variables with which data could be merged across missions for future analyses. Data were merged across multiple missions in order to inform analyses for the development of shorter-forms of some measures and to inform projects such as the HFBP-EM harmonization project. Second, the BHP Laboratory assessed the reliability and validity of the standard-length surveys included in the HFBP-EM suite and identified key measurement gaps across instruments. The results of these analyses were used to develop shorter versions of several measures in the exploration suite. The short form survey battery is intended to decrease participant burden, improve spaceflight operational acceptability, and improve behavioral medicine and teams risk assessment. Subject Matter Expert (SME) input was also collected for the short forms to ensure content coverage and to provide additional validity evidence. The short form surveys are being validated in the HERA Campaign 6. Data collection is underway. Finally, we worked with the Research Operations and Integration (ROI) element to prepare for HERA C6 missions and collected data on the protocol for one mission. The initial start date for Campaign 6 Mission 1 was delayed until 09/16/2021 due to the Covid19 pandemic. Campaign 6 mission 1 was completed on 11/22/2021, n=4 participants. Objective performance data yield (Cognition and ROBoT): 99.46%, Wearables Data yield (Actigraphy and Heart rate): 84.13%; Survey data yield is still being assessed. Wearables data yield was reduced in this mission relative to previous HERA campaigns due to significant skin irritation and possible allergic reaction from electrode patches linked to the heart rate devices. Alternative options are being explored for the next mission. In 2022, we will continue to collect data on the remaining three Campaign 6 missions, process and prepare data. We will also focus on getting data shared with external investigators to these research teams in a timely manner, so they can determine the efficacy of their countermeasures.
Bibliography Type:	Description: (Last Updated: 08/01/2025)
Articles in Peer-reviewed Journals	Marcinkowski MA, Bell ST, Roma PG. "The nature of conflict for teams in isolated, confined, and extreme environments." Acta Astronautica. 2021 Apr;181:81-91. https://doi.org/10.1016/j.actaastro.2021.01.004 , Apr-2021
Articles in Peer-reviewed Journals	Smith MG, Wusk GC, Nasrini J, Baskin P, Dinges DF, Roma PG, Basner M. "Effects of six weeks of chronic sleep restriction with weekend recovery on cognitive performance and wellbeing in high-performing adults." Sleep. 2021 Aug 13;44(8):zsab051. https://doi.org/10.1093/sleep/zsab051 ; PMID: 33630069 , Aug-2021